Math 2551 Worksheet: Cross Products

1. Let $P = (1, -1, 2)$, $Q = (2, 0, -1)$, and $R = (0, 2, 1)$.
   (a) Find the area of the triangle determined by the points P, Q, and R.
   (b) Find a unit vector normal to the plane containing P, Q, and R.
2. If the statement is *always* true, answer true. If the statement is *ever* false, answer false. Justify your answer.

   In each case, $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in $\mathbb{R}^3$.
   (a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
   (b) $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^{\mathbf{2}}$
   (c) $(\mathbf{u} \times \mathbf{u}) \cdot \mathbf{u} = \mathbf{0}$
3. Suppose $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are vectors in $\mathbb{R}^3$. Which of the following make sense, and which do not? For those that make sense, is the result a vector or a scalar?
   (a) $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$
   (b) $\mathbf{u} \times (\mathbf{v} \cdot \mathbf{w})$
   (c) $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$
   (d) $\mathbf{u} \cdot (\mathbf{v} \cdot \mathbf{w})$
4. Let $\mathbf{u} = \langle 2, 3 \rangle$. Find the maximum possible value for $\mathbf{u} \cdot \mathbf{v}$ if $\mathbf{v}$ is a unit vector, and find a $\mathbf{v}$ which gives this maximum. Then repeat the problem with “maximum” replaced by “minimum.”